CS383 · SOFTWARE ENGINEERING

Midterm 462 Practice

20 marks

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01 Questions

Q1 MCQ · 2 M

Choose the correct answer, then fill up the table with the correct choice:

1. What does product quality depend on?

   a. Development process b. Developer skills c. Project size d. All of the above

2. What is one of the process attributes that focuses on how easy it is to understand the process definition?

   a. Standardization b. Understandability c. Visibility d. Measurability

3. Which of the following is a key principle of Extreme Programming (XP)?

   a. Long-term planning b. Continuous feedback c. Heavy documentation d. Heavy documentation

4. In Extreme Programming (XP), what is the practice of continuously improving the code[base]?

   a. Refactoring b. Pair Programming c. Test Automation d. Continuous Integration

Q2 ETHICS · 8 M

According to the 8 principles of the IEEE Code of Ethics (PUBLIC, CLIENT AND EMPLOYER, PRODUCT, JUDGMENT, MANAGEMENT, PROFESSION, COLLEAGUES, SELF), identify the type of ethics reflected in the following descriptions:

SEEPP ethic example	Ethics Principle
Give a fair hearing to the opinions, concerns, or complaints of a colleague.	
Improve their knowledge of this Code, its interpretation, and its application to their work.	
Ensure that software engineers are informed of standards before being held to them.	
Ensure an appropriate method is used for any project on which they work or propose to work.	
Disclose to all concerned parties those conflicts of interest that cannot reasonably be avoided or escaped.	
Maintain the integrity of data, being sensitive to outdated or flawed occurrences.	
Accept full responsibility for their own work.	
Not give unfair treatment to anyone because of any irrelevant prejudices.	

Q3 FR AND NFR · 4 M

Extract 2 functional requirements and 2 non-functional requirements from the following software description.

A company needs a Task Management System to streamline its internal processes. The system will allow employees to create tasks with a title, description, priority level, and due date, and assign these tasks to [team] members. Employees can update the status of their tasks as they work on them. The system should handle [up to] 100 concurrent users without delays and ensure that user data is encrypted during transmission.

Q4 FOUR ACTIVITIES · 2 M

What are the 4 main software engineering activities. The activities must be given in a logical order to ensure the successful delivery of software products:

Q5.1 DEVELOPMENT MODEL · 4 M

Select the most appropriate software development model (Plan-Driven, Iterative, or Agile) based on key project characteristics. Justify your choice.

1. Project 1 Description:

• Stakeholder Presence: Limited. Stakeholders provide initial requirements but are not available for frequent feedback.

• Requirement Volatility: Low. Requirements are well-defined and unlikely to change.

• Project Size: Large-scale project with multiple teams working in parallel.

• Team Structure: Teams are geographically distributed, and communication is primarily through documentation.

Q5.2 PROJECT 2

2. Project 2 Description:

• Stakeholder Presence: High. Stakeholders are actively involved and provi[de continuous feedback].

• Requirement Volatility: High. Requirements are expected to chang[e as the project] progresses.

• Project Size: Small team of 5 developers working on a web applicat[ion].

• Team Structure: Co-located team with daily stand-up meetings.

Q5.3 PROJECT 3

3. Project 3 Description:

• Stakeholder Presence: Moderate. Stakeholders are available for periodic reviews but not continuously.

• Requirement Volatility: Moderate. Some changes are expected, but the core requirements are stable.

• Project Size: Medium-sized project with a team of 15 developers.

• Team Structure: The team is co-located, and there is a need for regular iterations to refine the product.

Q5.4 PROJECT 4

4. Project 4 Description:

• Stakeholder Presence: Limited. Stakeholders provide initial requirements and review progress at major milestones.

• Requirement Volatility: Low. Requirements are stable and well-documented.

• Project Size: Medium-sized project with a focus on regulatory compliance.

• Team Structure: Teams are experienced and follow a structured process with clear documentation.